1. UNDERSTANDING SYNCHRONIZATION IN PLAIN ENGLISH

Core Objective: Record an expense or income on any device (your desktop PC, your smartphone, or your wife's tablet) without **EVER** having to retype it somewhere else. All devices stay continuously balanced down to the exact cent.

The "Smart Merge" Guarantee (Zero Overwrites & Zero Data Loss)

A major concern in computing is that one device might accidentally overwrite or erase work done on another. In Rachel Eagles, this risk is **technically impossible** thanks to non-destructive smart merging:

- If your PC contains entry **A** and your phone records entry **B**, syncing merges them into **A + B**. Both devices gain the full updated ledger.
- If you capture a member's picture with your smartphone camera, the photo transfers to the PC and displays inside the official desktop directory.
- The 7 official expenditure families always stay strictly arranged in alphabetical order.

2. DUAL SYNCHRONIZATION CHANNELS: LOCAL WI-FI VS 5G MOBILE CLOUD

MODE	WHEN TO USE IT?	HOW IT WORKS IN PRACTICE
LOCAL WI-FI	At home or office when PC and mobile share the same Wi-Fi router.	Direct device-to-device communication on port 8085. Blazing fast (0.2s), secure, and does not require external internet.
5G MOBILE CLOUD	On the go, during meetings, group charity events, or abroad.	Phone pushes encrypted records to your secure Cloud vault (GitHub Contents API). Whenever the PC starts up, it pulls all updates.
SMART SYNC (AUTO)	**Recommended Default**	Tests in 800ms whether the home PC is reachable over Wi-Fi. If yes, syncs locally; if away on cellular data, routes via 5G Cloud automatically!

3. CONCRETE REAL-WORLD SCENARIOS

1. **Outdoor Scenario (Smartphone over 5G):** You are attending a Rachel Eagles assembly. A brother hands you **$250 cash** for the *Community Service* fund. You open the app on your phone, click + *New Credit*, type $250 and tap 🔄. In the evening at home, you turn on your PC: the $250 entry is already logged in the general ledger!

2. **Office Scenario (Home PC):** You review 8 utility and supplies receipts using your widescreen monitor, keyboard, and mouse. You click 🔄 **Sync** in the top header. When your wife turns on her tablet or when you pick up your phone, all 8 receipts appear instantly with the recalculation of the net balance.

3. **Member Picture Scenario:** You take a portrait of a new member using your smartphone camera in the *Members* tab. Once synced, their official photo—compressed into an optimized 280x280 square—shows on the PC ready for printing the certified membership booklet.

4. THE TOP HEADER BAR (ACCESSIBLE FROM EVERYWHERE)

BUTTON / ITEM	VISUAL REPRESENTATION	ROLE & OPERATION
Quick Sync Button	Sync	**Most essential button:** triggers instant bidirectional synchronization. A colored status dot indicates connection health: • Green = Connected and up to date. • Blue = Sync currently in progress. • Red = Offline (prompts 5G Cloud fallback).
Language Switcher	EN / FR	Instantly switches the complete user interface, form labels, charts, and date picker between English and French.
Info Button	i INFO	Displays the official group identification panel (foundation date, active roster counts, currency settings, app version).
Logout / Exit Button	▯	Allows secure session locking or clean application shutdown.

5. THE 8 MAIN NAVIGATION TABS & ACTION BUTTONS

1. Home Tab (Dashboard & Balances)

BUTTON	COLOR	ACTION DESCRIPTION
+ New Debit	+ New Debit	Opens quick form to record an expenditure with category, date, payee and amount.
+ New Credit	+ New Credit	Opens quick form to log an income, dues payment or donation received.
View Analytics	Analytics	Jumps directly to visual spending trends and category breakdown charts.
Share Shortcuts	WhatsApp / Email	Generates a formatted treasury summary ready to forward to executive officers.

2. Debits & Credits Tabs (General Ledger)

BUTTON	COLOR	ACTION DESCRIPTION
Date Range Filter	Calendar	Filters ledger entries across custom periods with bilingual date pickers.
Export to PDF	Export PDF	Generates official A4 certified ledger sheets with crest, eagle seal watermark and totals.
Edit / Delete		Available on every row to adjust amounts, payees or remove erroneous records.

3. Members Tab (Directory & Portrait Photos)

BUTTON	APPEARANCE	FUNCTIONAL PURPOSE
Add Member	+ Add Member	Opens profile modal to register name, group rank, phone, email, profession and emergency contact.
Cards / Table Switcher	Cards / Table	Toggles between visual photo cards (ideal for tablets) and high-density spreadsheet view (ideal for PC).
Select Member Photo	Photo	Opens smartphone camera or PC file explorer. Auto-crops and compresses into 280x280 square JPEG to keep database light.
Export Members PDF	Directory PDF	Prints certified membership roster booklet with color portraits and group titles.

4. Analytics, Reports & Audit Tab

BUTTON	APPEARANCE	FUNCTIONAL PURPOSE
Official Audit Report	Export Audit PDF	Generates time-phased audit roadmaps with 6 action cards (0-30d, 30-60d, 60-90d).
3-Month Forecast Report	Export Forecast	Projects cash flow requirements and liquidity safety targets.
Interactive Action Badges	TO PLAN / COMPLETED	Clicking toggles progress status. State is persistently synced across all devices.

5. PC Sync & Backup Tab (Offline Operations)

BUTTON	APPEARANCE	FUNCTIONAL PURPOSE
Download Complete Backup (.json)	Backup (.json)	Exports 100% self-contained file with all transactions, members and photos. Perfect for USB flash drives.
Restore / Merge (.json)	Restore	Safely imports and merges offline backups without needing any Wi-Fi or cellular network.
Generate ZIP Archive	ZIP Archive	Creates compressed package with CSV and transactions filtered by chosen date range.

6. TROUBLESHOOTING & FREQUENTLY ASKED QUESTIONS

1. What if local Wi-Fi sync reports a connection error?
Ensure your PC and phone are connected to the same local Wi-Fi router. If you are away from home, simply switch to **5G Cloud** mode with one tap.

2. How to transfer data if there is NO network at all (no Wi-Fi, no 5G)?
Navigate to *PC Sync & Backup*, click *Download Complete Backup (.json)*. Copy this small file to a USB thumb drive or transfer it via cable to the PC, then click *Restore / Merge*. All records merge seamlessly!

3. Are member photos preserved during synchronization?
Yes, 100%. All compressed photos are embedded directly into both the JSON backups and the 5G Cloud vault.